

Question

Write a Java program to demonstrate hybrid inheritance using classes and interfaces. Calculate the area and perimeter of a rectangle using inheritance.

Steps

1. Create a class `Shape`.
2. Create a method `display()` in the `Shape` class.
3. Create an interface `Area` with the method `calculateArea()`.
4. Create another interface `Perimeter` with the method `calculatePerimeter()`.
5. Create a class `Rectangle` that extends `Shape` and implements both `Area` and `Perimeter`.
6. Create a constructor in `Rectangle` to initialize length and breadth.
7. Override the `display()` method from the parent class.
8. Implement `calculateArea()` to calculate:
 $\text{Area} = \text{length} \times \text{breadth}$
9. Implement `calculatePerimeter()` to calculate:
 $\text{Perimeter} = 2 \times (\text{length} + \text{breadth})$
10. Overload a `calculate()` method to demonstrate method overloading.
11. Create a `Rectangle` object in the `main()` method.
12. Display the area and perimeter.

Algorithm

1. **Start**
2. Define the **Shape** class.
3. Define the **Area** interface.
4. Define the **Perimeter** interface.
5. Create the **Rectangle** class using:

class Rectangle extends Shape implements Area, Perimeter

6. Initialize length and breadth using the constructor.
7. Override the **display()** method.
8. Calculate the area using:

Area = length × breadth

9. Calculate the perimeter using:

Perimeter = 2 × (length + breadth)

10. Define an overloaded **calculate()** method.
11. Create a **Rectangle** object with length **10** and breadth **5**.
12. Call the required methods.
13. Display the results.
14. **Stop**

```
id_80caa912\bin' 'HybridInheritance'  
Rectangle  
Area: 50  
Perimeter: 30  
Length: 10  
Area: 50  
PS C:\Users\venik\OneDrive\Assignmets\Assignment1\Hybrid>
```